Security Analysis of Haldiram's Web Infrastructure Using Nmap

Introduction

With the rapid growth of e-commerce platforms, website security has become a critical requirement for protecting customer data and online services. This project focuses on performing a security analysis of the Haldiram's website infrastructure using the network scanning tool Nmap.

The goal is to identify open ports, running services, and possible vulnerabilities in the website infrastructure. By performing vulnerability scanning and analysing the results, potential security weaknesses can be identified and mitigation strategies can be recommended.

Objectives

The main objectives of this project are:

- Identify open ports on the Haldi ram's web server.
- Detect services running on the server.
- Perform vulnerability scanning using Nmap scripts.
- Analyze potential security weaknesses.
- Provide recommendations to improve website security.

Tools Used

Tool	Purpose
Nmap	Network scanning and vulnerability detection
Operating System	Kali Linux / Linux
Terminal	Command execution
Web Browser	https://www.haldirams.com

Nmap

Nmap (Network Mapper) is an open-source tool used for network discovery and security auditing. It helps detect open ports, operating systems, services, and vulnerabilities.

Methodology

The security analysis follows these steps:

- Target Identification
- Host Discovery
- Port Scanning
- Service Detection
- Vulnerability Detection using NSE Scripts
- Result Analysis
- Security Recommendations

Target Information

Target Website

https://www.haldirams.com

Basic Information

Parameter	Value
Domain	haldirams.com
Server Location	India
Web Server	awselb (Amazon Load Balancer)
Domain Registered	1999

The website infrastructure appears to use cloud hosting and load balancing technologies for high availability.

Nmap Scanning Commands

Host Discovery

Command

nmap haldirams.com

Purpose

Checks if the host is online and performs a basic scan.

Port Scan

Command

nmap -p- haldirams.com

Purpose

Scans all 65535 ports to detect open ports.

Service Version Detection

Command

nmap -sV haldirams.com

Purpose

Identifies services and software versions running on the server.

Results

Port	Service
80	HTTP
443	HTTPS

OS Detection

Command

nmap -O haldirams.com

Purpose

Attempts to detect the operating system used by the server.

Vulnerability Scan using NSE Scripts

Command

nmap --script vuln haldirams.com

Purpose

Runs Nmap vulnerability scripts to detect:

- Security misconfigurations
- Known vulnerabilities
- Weak SSL/TLS configurations

Possible Vulnerabilities Identified

During vulnerability analysis, the following types of issues may appear:

1. Open Ports

   Unnecessary open ports may allow attackers to access services.

2. Outdated Services

   Older software versions may contain known vulnerabilities.

3. Weak SSL/TLS Configuration

Improper encryption may expose sensitive data.

4. Missing Security Headers

Lack of HTTP security headers may allow attacks such as:

Cross-Site Scripting (XSS)

Clickjacking

Data injection

Risk Analysis

Vulnerability	Risk Level	Impact
Open Ports	Medium	Unauthorized access
Outdated Services	High	Exploitation by attackers
Weak Encryption	High	Data leakage
Missing Headers	Medium	Web attacks

Security Recommendations

To improve website security, the following measures are recommended:

- Close unused ports
- Update server software regularly
- Implement strong SSL/TLS encryption
- Enable security headers such as:
- Content Security Policy
- HSTS
- X-Frame-Options
- Implement Web Application Firewall (WAF)
- Perform regular vulnerability scanning
- Use intrusion detection systems

Conclusion

This project demonstrated how Nmap can be used to analyze the security posture of a web infrastructure. By scanning the Haldiram's website, open ports, running services, and potential vulnerabilities can be identified.

The results highlight the importance of regular security assessments and proactive vulnerability management to protect web applications and user data from cyber threats.

Screenshots

```
root@kali: /home/kali
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo su
┌──(root㉿kali)-[/home/kali]
└─# sudo apt-get update && sudo apt-get upgrade -y
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Reading package lists ... Done
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
Calculating upgrade ... Done
The following packages were automatically installed and are no longer required:
  libann0 libaudio2 libcdt5 libcgraph6 libcrypt-dev libdlt2 libgdk-pixbuf2.0-bin libio-pty-perl libipc-run-perl liblab-gamut1 libmupdf25.1 libplacebo351
  librubberband2 libsframe2 libsimdjson29 libsimdutf29 libsnmp40t64 libtsk19t64 libtss2-fapi1t64 libzxing3 postgresql-common-dev python3-aiocache
  python3-aiomcache python3-pysnmp4 python3-terminaltables python3-wapiti-arsenic python3-yaswfp ruby-unf-ext
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.

┌──(root㉿kali)-[/home/kali]
└─# nmap www.haldirams.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-09 21:46 +0530
Nmap scan report for www.haldirams.com (151.101.157.55)
Host is up (0.024s latency).
Other addresses for www.haldirams.com (not scanned): 64:ff9b::9765:9d37
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 5.42 seconds
```

```
root@kali: /home/kali
Session  Actions  Edit  View  Help
Nmap done: 1 IP address (1 host up) scanned in 5.42 seconds

┌──(root㉿kali)-[/home/kali]
└─# nmap -sV -A -script=vuln 151.101.157.55 -oA Haldirams_Security_Analysis_report
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-09 21:54 +0530
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 0 undergoing Script Pre-Scan
NSE Timing: About 0.00% done
Stats: 0:05:33 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 97.92% done; ETC: 21:59 (0:00:06 remaining)
Nmap scan report for 151.101.157.55
Host is up (0.034s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE    VERSION
80/tcp  open  http-proxy Varnish
| http-slowloris-check:
|   VULNERABLE:
|   Slowloris DOS attack
|     State: LIKELY VULNERABLE
|     IDs:  CVE:CVE-2007-6750
|       Slowloris tries to keep many connections to the target web server open and hold
|       them open as long as possible.  It accomplishes this by opening connections to
|       the target web server and sending a partial request. By doing so, it starves
|       the http server's resources causing Denial Of Service.
|
|     Disclosure date: 2009-09-17
|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|_      http://ha.ckers.org/slowloris/
| http-phpmyadmin-dir-traversal:
|   VULNERABLE:
|   phpMyAdmin grab_globals.lib.php subform Parameter Traversal Local File Inclusion
|     State: LIKELY VULNERABLE
|     IDs:  CVE:CVE-2005-3299
|       PHP file inclusion vulnerability in grab_globals.lib.php in phpMyAdmin 2.6.4 and 2.6.4-pl1 allows remote attackers to include local files via the $__
redirect parameter, possibly involving the subform array.
|
|     Disclosure date: 2005-10-nil
|     Extra information:
|       ../../../../etc/passwd not found.
|
|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2005-3299
|_      http://www.exploit-db.com/exploits/1244/
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
443/tcp open  ssl/https
|_http-csrf: Couldn't find any CSRF vulnerabilities.
| http-slowloris-check:
|   VULNERABLE:
|   Slowloris DOS attack
```

```
root@kali: /home/kali
Session  Actions  Edit  View  Help
SF:content-type:\x20text/plain;\x20charset=utf-8\r\nx-served-by:\x20cache-
SF:sxv4660022\r\n\r\nRequested\x20host\x20does\x20not\x20match\x20any\x20S
SF:ubject\x20Alternative\x20Names\x20\(SANs\)\x20on\x20TLS\x20certificate\
SF:x20\[e4ca160dcee5cc7ec7120b704932efaa70af72fb86eb6bd0ee2282bc84f05d07\]
SF:\x20in\x20use\x20with\x20this\x20connection\.\r\n\r\nVisit\x20https://w
SF:ww\.fastly\.com/documentation/guides/concepts/errors/#routing-errors\x2
SF:0for\x20more\x20information\.\r\n\r")%r(tor-versions,96,"HTTP/1\.1\x204
SF:00\x20Bad\x20Request\r\nConnection:\x20close\r\nContent-Length:\x2011\r
SF:\ncontent-type:\x20text/plain;\x20charset=utf-8\r\nx-served-by:\x20cach
SF:e-sxv4660021\r\n\r\nBad\x20Request")%r(RTSPRequest,96,"HTTP/1\.1\x20400
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SF:content-type:\x20text/plain;\x20charset=utf-8\r\nx-served-by:\x20cache-
SF:sxv4660021\r\n\r\nBad\x20Request");
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results incomplete
No OS matches for host
Network Distance: 5 hops

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1   7.34 ms  10.101.252.74
2   ...
3   45.91 ms 255.0.0.2
4   45.76 ms 255.0.0.3
5   45.63 ms 151.101.157.55

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 556.86 seconds

┌──(root㉿kali)-[/home/kali]
└─# xsltproc Haldirams_Security_Analysis_report.xml -o Haldirams_Security_Analysis_report.html
```

```
Session  Actions  Edit  View  Help
Nmap done: 1 IP address (1 host up) scanned in 5.42 seconds

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Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-09 21:54 +0530
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Nmap scan report for 151.101.157.55
Host is up (0.034s latency).
Not shown: 998 filtered tcp ports (no-response)
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|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|_      http://ha.ckers.org/slowloris/
| http-phpmyadmin-dir-traversal:
|   VULNERABLE:
|   phpMyAdmin grab_globals.lib.php subform Parameter Traversal Local File Inclusion
|     State: LIKELY VULNERABLE
|     IDs:  CVE:CVE-2005-3299
|       PHP file inclusion vulnerability in grab_globals.lib.php in phpMyAdmin 2.6.4 and 2.6.4-pl1 allows remote attackers to include local files via the $__
redirect parameter, possibly involving the subform array.
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|     Disclosure date: 2005-10-nil
|     Extra information:
|       ../../../../etc/passwd not found.
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|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
443/tcp open  ssl/https
|_http-csrf: Couldn't find any CSRF vulnerabilities.
| http-slowloris-check:
|   VULNERABLE:
|   Slowloris DOS attack
|     State: LIKELY VULNERABLE
|     IDs:  CVE:CVE-2007-6750
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|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|_      http://ha.ckers.org/slowloris/
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
| fingerprint-strings:
|   FourOhFourRequest:
|     HTTP/1.1 421 Misdirected Request
|     Connection: close
|     Content-Length: 291
|     content-type: text/plain; charset=utf-8
|     x-served-by: cache-sxv4660022
|     Requested host does not match any Subject Alternative Names (SANs) on TLS certificate [e4ca160dcee5cc7ec7120b704932efaa70af72fb86eb6bd0ee2282bc84f05d07
] in use with this connection.
|     Visit https://www.fastly.com/documentation/guides/concepts/errors/#routing-errors for more information.
|   GetRequest, HTTPOptions:
|     HTTP/1.1 421 Misdirected Request
|     Connection: close
|     Content-Length: 291
|     content-type: text/plain; charset=utf-8
|     x-served-by: cache-sxv4660021
|     Requested host does not match any Subject Alternative Names (SANs) on TLS certificate [e4ca160dcee5cc7ec7120b704932efaa70af72fb86eb6bd0ee2282bc84f05d07
] in use with this connection.
|     Visit https://www.fastly.com/documentation/guides/concepts/errors/#routing-errors for more information.
|   RTSPRequest, tor-versions:
|     HTTP/1.1 400 Bad Request
|     Connection: close
|     Content-Length: 11
|     content-type: text/plain; charset=utf-8
|     x-served-by: cache-sxv4660021
|_    Request
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cg
i?new-service :
SF-Port443-TCP:V=7.98%T=SSL%I=7%D=4/9%Time=69D7D2C6%P=x86_64-pc-linux-gnu%
SF:r(GetRequest,1B7,"HTTP/1\.1\x20421\x20Misdirected\x20Request\r\nConnect
SF:ion:\x20close\r\nContent-Length:\x20291\r\ncontent-type:\x20text/plain;
SF:\x20charset=utf-8\r\nx-served-by:\x20cache-sxv4660021\r\n\r\nRequested\
SF:x20host\x20does\x20not\x20match\x20any\x20Subject\x20Alternative\x20Nam
SF:es\x20\(SANs\)\x20on\x20TLS\x20certificate\x20\[e4ca160dcee5cc7ec7120b7
SF:04932efaa70af72fb86eb6bd0ee2282bc84f05d07\]\x20in\x20use\x20with\x20thi
SF:s\x20connection\.\r\n\r\nVisit\x20https://www\.fastly\.com/documentatio
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SF:ubject\x20Alternative\x20Names\x20\(SANs\)\x20on\x20TLS\x20certificate\
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└─#
```